⚙ Fluid Power Systems — Unit Conversions Reference

Course: MEC 431 — BUC | **Prepared by:** Eng. Moh Zedan (TA)

1. Pressure

From	To	Multiply By	Example
bar	Pa	$\times 10^5$	1 bar = 100,000 Pa
bar	kPa	$\times 100$	1 bar = 100 kPa
bar	MPa	$\times 0.1$	10 bar = 1 MPa
bar	**psi**	$\times 14.5$	**1 bar = 14.5 psi**
psi	**bar**	$\times 0.069$	**1 psi = 0.069 bar**
psi	kPa	$\times 6.895$	1 psi = 6.895 kPa
kPa	psi	$\times 0.145$	1 kPa = 0.145 psi
atm	bar	$\times 1.01325$	1 atm ≈ 1.013 bar
atm	psi	$\times 14.696$	1 atm = 14.696 psi
atm	Pa	$\times 101325$	1 atm = 101,325 Pa
mmHg (Torr)	Pa	$\times 133.322$	760 mmHg = 1 atm

Quick Memory Aid — Pressure

$1 \text{ bar} \approx 100 \text{ kPa} \approx 14.5 \text{ psi} \approx 1 \text{ atm}$

2. Flow Rate (Volume per Time)

From	To	Multiply By	Example
L/min	**m³/s**	$\times \frac{10^{-3}}{60} = \times 1.667 \times 10^{-5}$	**60 L/min = 0.001 m³/s**
m³/s	**L/min**	$\times 60000$	**0.001 m³/s = 60 L/min**
L/min	m³/min	$\times 10^{-3}$	39 L/min = 0.039 m³/min
GPM (US)	L/min	$\times 3.785$	1 GPM = 3.785 L/min
L/min	GPM (US)	$\times 0.264$	1 L/min = 0.264 GPM
GPM	in³/s	$\times 3.85$	—

m^3/s	GPM	$\times 15850.3$	—

Key Conversion Step (used in every problem):

$$Q\ [\mathrm{m^3/s}] = \frac{Q\ [\mathrm{L/min}]}{1000 \times 60} = \frac{Q\ [\mathrm{L/min}]}{60000}$$

3. Length

From	To	Multiply By	Example
mm	**m**	$\times 10^{-3}$	**90 mm = 0.09 m**
cm	m	$\times 10^{-2}$	25 cm = 0.25 m
in	**m**	$\times 0.0254$	**1 in = 0.0254 m**
in	mm	$\times 25.4$	1 in = 25.4 mm
ft	m	$\times 0.3048$	1 ft = 0.3048 m
m	ft	$\times 3.281$	1 m = 3.281 ft

4. Area

From	To	Multiply By	Example
mm^2	**m^2**	$\times 10^{-6}$	**1000 mm^2 = 0.001 m^2**
cm^2	m^2	$\times 10^{-4}$	63.62 cm^2 = 6.362×10^{-3} m^2
in^2	**m^2**	$\times 6.452 \times 10^{-4}$	**1 in^2 = 6.452×10^{-4} m^2**
in^2	cm^2	$\times 6.452$	1 in^2 = 6.452 cm^2
m^2	in^2	$\times 1550$	—

Circle Area (most used in Fluid Power):

$A = \frac{\pi}{4} d^2$ — always convert diameter to **meters** first!

5. Force

From	To	Multiply By	Example
N	kN	$\times 10^{-3}$	1000 N = 1 kN
N	**lbf**	$\times 0.2248$	**1 N = 0.2248 lbf**
lbf	**N**	$\times 4.448$	**1 lbf = 4.448 N**
kgf	N	$\times 9.807$	1 kgf = 9.807 N
N	kgf	$\times 0.1020$	—

6. Torque

From	To	Multiply By	Example
N·m	**ft·lbf**	$\times 0.7376$	**1 N·m = 0.7376 ft·lbf**
ft·lbf	**N·m**	$\times 1.3558$	**1 ft·lbf = 1.3558 N·m**
N·m	in·lbf	$\times 8.851$	—

7. Power

From	To	Multiply By	Example
W	kW	$\times 10^{-3}$	12675 W = 12.675 kW
kW	**hp**	$\times 1.341$	**1 kW = 1.341 hp**
hp	**kW**	$\times 0.7457$	**1 hp = 0.7457 kW**
hp	W	$\times 745.7$	1 hp = 745.7 W
hp	ft·lbf/s	$\times 550$	1 hp = 550 ft·lbf/s

Hydraulic Power Formulas:

SI Units: $P_h = \frac{p \cdot Q}{60000}$ [kW] where p in kPa, Q in L/min

OR: $P_h = p \cdot Q$ [W] where p in Pa, Q in m^3/s

British Units: $P_h = \frac{p \cdot Q}{1714}$ [hp] where p in psi, Q in GPM

Mechanical Power Formulas:

SI: $P_m = \frac{T \cdot n}{9550}$ [kW] where T in N·m, n in RPM

British: $P_m = \frac{T \cdot n}{5252}$ [hp] where T in ft·lbf, n in RPM

8. Volume

From	To	Multiply By	Example
L	**m³**	$\times 10^{-3}$	**1 L = 0.001 m³**
mL	cm³ (cc)	$\times 1$ (same)	1 mL = 1 cm³
US gallon	L	$\times 3.785$	1 gal = 3.785 L
L	US gallon	$\times 0.2642$	1 L = 0.2642 gal
in³	cm³	$\times 16.387$	1 in³ = 16.387 cm³
in³	L	$\times 0.01639$	—
ft³	L	$\times 28.317$	1 ft³ = 28.317 L

9. Velocity

From	To	Multiply By	Example
m/s	**cm/s**	$\times 100$	**0.1022 m/s = 10.22 cm/s**
m/s	ft/s	$\times 3.281$	—
ft/s	m/s	$\times 0.3048$	—
m/s	in/s	$\times 39.37$	—
in/s	m/s	$\times 0.0254$	—

10. Mass & Density

From	To	Multiply By	Example
kg	lbm	$\times 2.205$	1 kg = 2.205 lbm
lbm	kg	$\times 0.4536$	1 lbm = 0.4536 kg
kg/m³	**lbm/ft³**	$\times 0.06243$	**900 kg/m³ = 56.2 lbm/ft³**
lbm/ft³	kg/m³	$\times 16.018$	—

Common Fluid Densities:
• Hydraulic Oil (mineral): $\rho \approx 850\text{–}900\ \text{kg/m}^3$
• Water: $\rho = 1000\ \text{kg/m}^3$
• Air (at STP): $\rho \approx 1.225\ \text{kg/m}^3$

11. Temperature

From	To	Formula	Example
°C	°F	$^\circ F = \frac{9}{5} \times {}^\circ C + 32$	50°C = 122°F
°F	°C	$^\circ C = \frac{5}{9}({}^\circ F - 32)$	212°F = 100°C
°C	K	$K = {}^\circ C + 273.15$	25°C = 298.15 K

12. Energy

From	To	Multiply By
J	kJ	$\times 10^{-3}$
kJ	BTU	$\times 0.9478$
BTU	kJ	$\times 1.055$
ft·lbf	J	$\times 1.3558$